TE1 Worked Examples: Trigonometric Identities

Recording companion for the TE1 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Which angles from the unit circle can I build this from, and what sign does the answer need? An angle in twelfths, like $7\pi/12$, is a sum or difference of angles you know, like $\pi/3 + \pi/4$. An angle in eighths, like $\pi/8$, is half of one, like $\pi/4$. A squared sine or cosine trades its square for a doubled angle. And every exact value has a sign that comes from the quadrant, so decide the sign before or after the formula, never by guessing.

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Example 1: Build 1

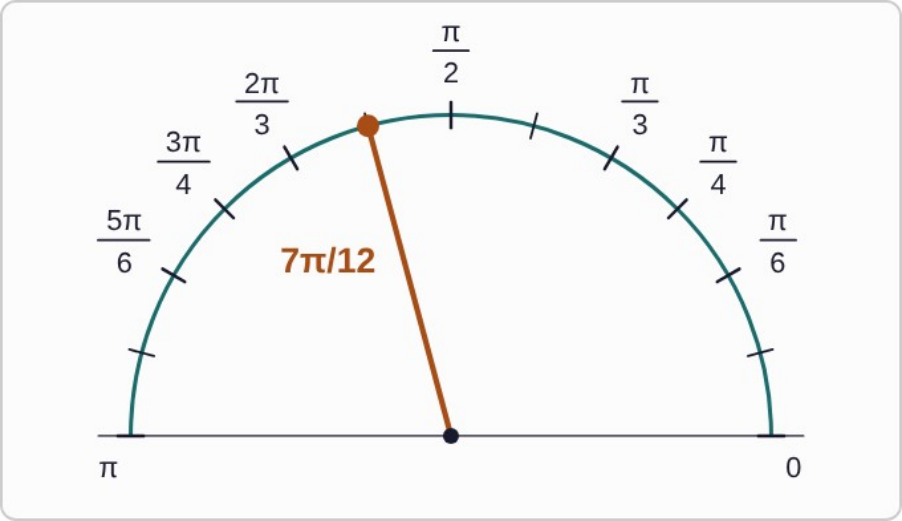


Figure: The upper half of the unit circle with a tick mark every pi/12, or 15 degrees. The ticks for pi/6, pi/4, pi/3, pi/2, 2pi/3, 3pi/4, and 5pi/6 are labeled. The angle 7pi/12 is drawn as a highlighted ray from the center, one tick past pi/2.

$$7\pi/12$$

Question

Which way of writing this angle uses only special angles?

- **A.** $\pi/6+\pi/4$
- **B.** $\pi/3+\pi/4$
- **C.** $\pi/2+\pi/12$
- **D.** $\pi/3-\pi/4$

Answer and reasoning

In twelfths, $\pi/3=4\pi/12$ and $\pi/4=3\pi/12$, so $\pi/3+\pi/4=7\pi/12$. The sum $\pi/6+\pi/4$ is only $5\pi/12$, and $\pi/3-\pi/4$ is $\pi/12$. The sum $\pi/2+\pi/12$ does equal $7\pi/12$, but $\pi/12$ is not one of the angles you know, so it does not help. **Use $\pi/3+\pi/4$.**

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Example 2: Build 2

$$\pi/12$$

Question

Which difference of special angles equals this angle?

A. $\pi/4-\pi/3$
B. $\pi/2-\pi/3$
C. $\pi/3-\pi/6$
D. $\pi/3-\pi/4$

Answer and reasoning

$\pi/3-\pi/4=4\pi/12-3\pi/12=\pi/12$. Order matters: $\pi/4-\pi/3=-\pi/12$. The other two both equal $\pi/6$. Another route also works: $\pi/12$ is half of $\pi/6$. ***$\pi/12=\pi/3-\pi/4$*.**

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Example 3: Build 3

$$\pi/8$$

Question

No sum or difference of special angles makes this angle. It is half of which special angle?

- **A.** $\pi/4$
- **B.** $\pi/16$
- **C.** $\pi/2$
- **D.** $\pi/6$

Answer and reasoning

Doubling $\pi/8$ gives $2\pi/8=\pi/4$. So $\pi/8$ is half of $\pi/4$, and a half-angle formula can use $\cos(\pi/4)=(\sqrt{2})/2$. Half of $\pi/8$ is $\pi/16$, which goes the wrong way. **It is half of $\boldsymbol{\pi/4}$.** Every angle in eighths works this way: $3\pi/8$ is half of $3\pi/4$.

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Example 4: Build 4

$$\sin(-5\pi/8)$$

Question

Before calculating: where is this angle, and what sign will the answer have?

A. Quadrant IV, so the sine is below zero
B. Quadrant II, so the sine is above zero
C. Quadrant III, so the sine is below zero
D. Quadrant III, so the sine is above zero

Answer and reasoning

A negative angle turns clockwise from the positive x-axis. $-5\pi/8$ is $-112.5°$: past $-90°$ but not yet $-180°$, so it lands in Quadrant III. Sine is the y-coordinate, which is below zero there. Quadrant II is where $+5\pi/8$ lands. You can also use $\sin(-\theta)=-\sin\theta$: $\sin(5\pi/8)$ is above zero, so $\sin(-5\pi/8)$ is below zero. **Quadrant III, and the sine is below zero.** When a half-angle formula gives ±, this is how you choose.

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Example 5: Exact 1

$$\sin(\pi/12)$$

Question

What is the exact value?

A. $(\sqrt{6}+\sqrt{2})/4$
B. $(\sqrt{6}-\sqrt{2})/4$
C. $(\sqrt{3}-\sqrt{2})/2$
D. $(\sqrt{2}-\sqrt{6})/4$

Answer and reasoning

With $\pi/12=\pi/3-\pi/4$: $\sin(\pi/3)\cos(\pi/4)-\cos(\pi/3)\sin(\pi/4)=(\sqrt{3})/2\cdot(\sqrt{2})/2-1/2\cdot(\sqrt{2})/2=(\sqrt{6}-\sqrt{2})/4\approx 0.259$. The answer $(\sqrt{3}-\sqrt{2})/2$ is $\sin(\pi/3)-\sin(\pi/4)$, which is not how sine works. The plus version is the cosine, and the reversed version is below zero, but $\pi/12$ is in Quadrant I. ***$(\sqrt{6}-\sqrt{2})/4$*.**

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Example 6: Exact 2

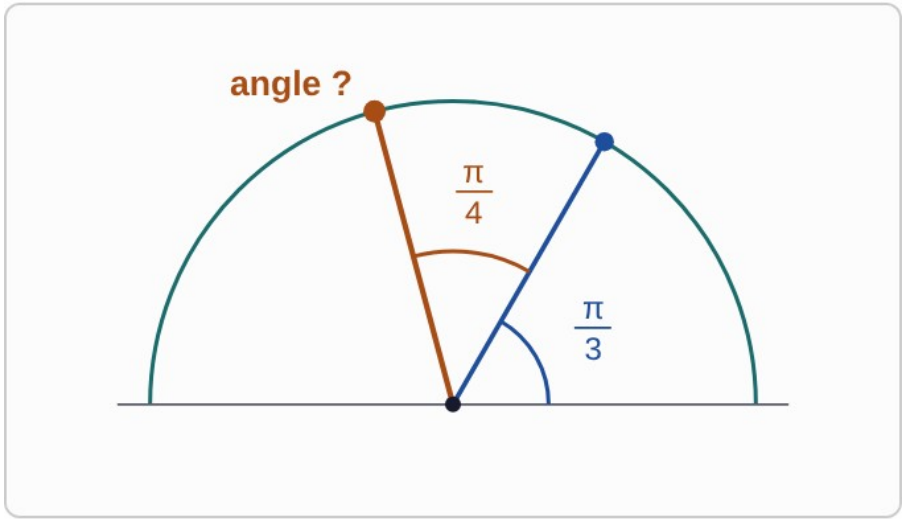


Figure: The upper half of the unit circle. A first ray makes an angle of pi/3 with the positive x-axis. A second ray is pi/4 farther counterclockwise, so the total angle from the positive x-axis to the second ray is pi/3 plus pi/4. The second ray is labeled angle question mark.

$$\cos(7\pi/12)$$

Question

What is the exact value?

- **A.** $(\sqrt{6}-\sqrt{2})/4$
- **B.** $(\sqrt{2}+\sqrt{6})/4$
- **C.** $(1+\sqrt{2})/2$
- **D.** $(\sqrt{2}-\sqrt{6})/4$

Answer and reasoning

With $7\pi/12=\pi/3+\pi/4$, the cosine formula flips to a minus: $\cos(\pi/3)\cos(\pi/4)-\sin(\pi/3)\sin(\pi/4)=(\sqrt{2})/4-(\sqrt{6})/4=(\sqrt{2}-\sqrt{6})/4\approx-0.259$. That makes sense: $7\pi/12$ is in Quadrant II, where cosine is below zero. The answer $(\sqrt{6}-\sqrt{2})/4$ has the right size but the wrong sign for Quadrant II. The plus version keeps the plus sign, and $(1+\sqrt{2})/2$ adds the two cosines, which is more than 1. ***$(\sqrt{2}-\sqrt{6})/4$.***

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Example 7: Exact 3

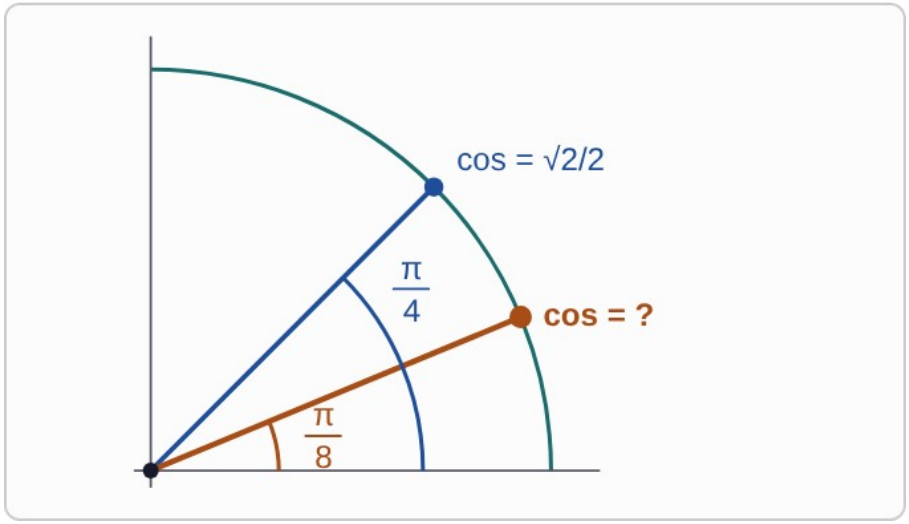


Figure: The first quadrant of the unit circle. A ray at pi/4 meets the circle at a point labeled cos = square root of 2 over 2. A second ray, halfway between the positive x-axis and the first ray, is at pi/8. Its point on the circle is labeled cos = question mark.

$$\cos(\pi/8)$$

Question

What is the exact value?

- **A.** $(\sqrt{2})/4$
- **B.** $(\sqrt{(2-\sqrt{2})})/2$
- **C.** $(\sqrt{(2+\sqrt{2})})/2$
- **D.** $\sqrt{((1+\sqrt{2})/2)}$

Answer and reasoning

$\pi/8$ is half of $\pi/4$, and it is in Quadrant I, so take the positive root: $\cos(\pi/8)=\sqrt{((1+\cos(\pi/4))/2)}=\sqrt{((1+\sqrt{2}/2)/2)}=\sqrt{((2+\sqrt{2})/4)}=(\sqrt{(2+\sqrt{2})})/2\approx 0.924$. The answer $(\sqrt{2})/4$ halves the cosine instead of the angle. The minus version is $\sin(\pi/8)$. The last one uses 1 + √2 instead of 1 + √2/2, and it is more than 1, which no cosine can be. **$(\sqrt{(2+\sqrt{2})})/2$.**

Example 8: Exact 4

$$\sin\left(\frac{17\pi}{12}\right)$$

Question

What is the exact value?

- **A.** $-(\sqrt{6}+\sqrt{2})/4$
- **B.** $(\sqrt{6}+\sqrt{2})/4$
- **C.** $-(\sqrt{6}-\sqrt{2})/4$
- **D.** $-(\sqrt{3}+1)/2$

Answer and reasoning

$17\pi/12 = 5\pi/4 + \pi/6$, so $\sin(5\pi/4)\cos(\pi/6) + \cos(5\pi/4)\sin(\pi/6) = -(\sqrt{2})/2 \cdot (\sqrt{3})/2 - (\sqrt{2})/2 \cdot 1/2 = -(\sqrt{6}+\sqrt{2})/4 \approx -0.966$. Check the sign: $17\pi/12$ is in Quadrant III, where sine is below zero. The $\sqrt{6}-\sqrt{2}$ version puts a minus between the two terms, which is the difference formula, and $-(\sqrt{3}+1)/2$ is less than -1, which no sine can be. **$-(\sqrt{6}+\sqrt{2})/4$.**

Example 9: Identity 1

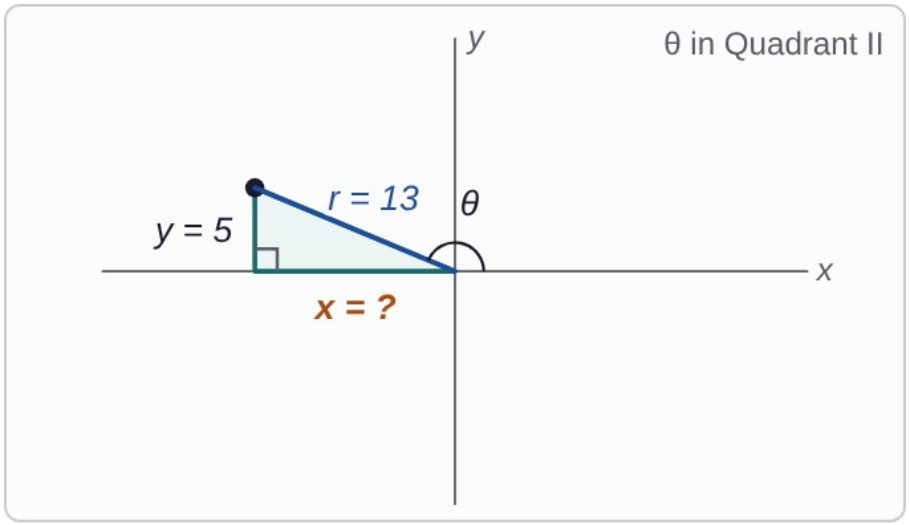


Figure: Coordinate axes with angle theta in Quadrant II. The terminal side goes from the origin up and to the left, with length r = 13. A vertical leg drops from its end to the x-axis and is labeled y = 5. The horizontal leg, along the x-axis from the origin to the left, is labeled x = question mark.

$$\sin\theta = 5/13$$

Question

Angle θ is in Quadrant II. What is $\cos\theta$?

- **A.** $12/13$
- **B.** $-8/13$
- **C.** $-12/5$
- **D.** $-12/13$

Answer and reasoning

$\cos^2\theta = 1-(5/13)^2 = 144/169$, so $\cos\theta = \pm 12/13$. In Quadrant II, x is to the left of the origin, so cosine is below zero. The answer $-8/13$ subtracts $13-5$ instead of squaring, and $-12/5$ is the cotangent, x/y. **$\cos\theta = -12/13$.**

Example 10: Identity 2

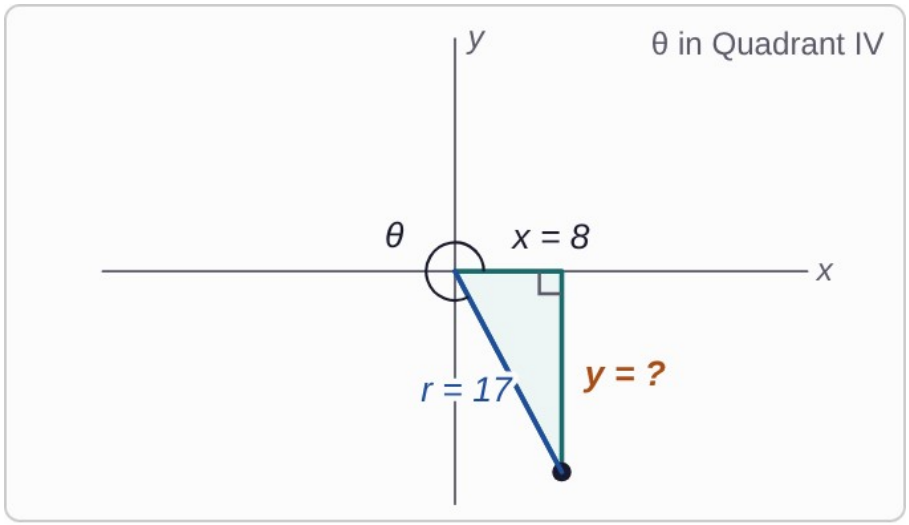


Figure: Coordinate axes with angle theta in Quadrant IV. The terminal side goes from the origin down and to the right, with length r = 17. The horizontal leg along the x-axis is labeled x = 8. The vertical leg, from the x-axis down to the end of the terminal side, is labeled y = question mark.

$$\cos\theta = 8/17$$

Question

Angle θ is in Quadrant IV. What is $\sin\theta$?

- **A.** $15/17$
- **B.** $-15/17$
- **C.** $-9/17$
- **D.** $-17/15$

Answer and reasoning

$\sin^2\theta = 1 - 64/289 = 225/289$, so $\sin\theta = \pm 15/17$. In Quadrant IV, y is below the origin, so sine is below zero. The answer $-9/17$ subtracts $17-8$, and $-17/15$ is the cosecant, which is less than -1, so it cannot be a sine. **$\sin\theta = -15/17$.**

Example 11: Identity 3

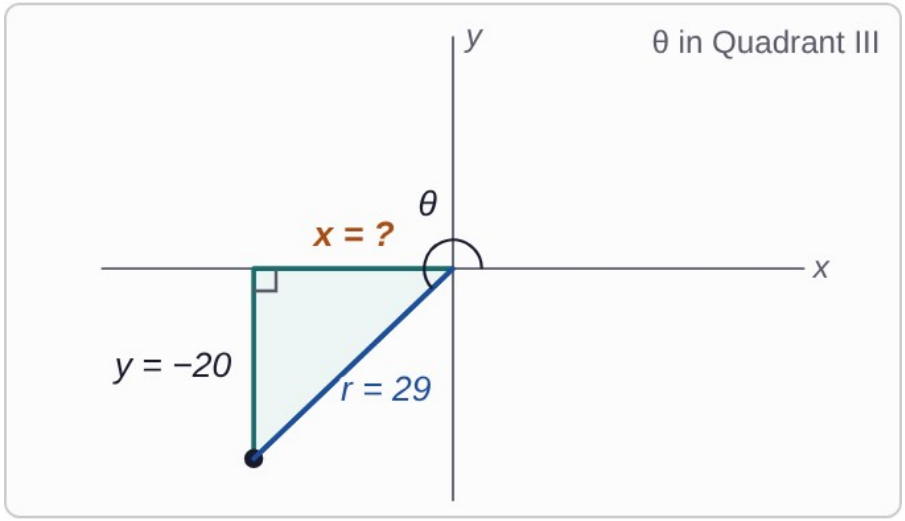


Figure: Coordinate axes with angle theta in Quadrant III. The terminal side goes from the origin down and to the left, with length r = 29. The vertical leg is labeled y = negative 20. The horizontal leg, along the x-axis to the left of the origin, is labeled x = question mark.

$$\sin\theta = -20/29$$

Question

Angle θ is in Quadrant III. What is $\cos\theta$?

- **A.** $21/29$
- **B.** $-9/29$
- **C.** $-21/29$
- **D.** $-20/21$

Answer and reasoning

$\cos^2\theta = 1 - 400/841 = 441/841$, so $\cos\theta = \pm 21/29$. In Quadrant III, both x and y are below zero. The answer $-9/29$ subtracts $29 - 20$, and $-20/21$ divides the two legs, y over x, with the sign flipped; that is not a cosine at all. **$\cos\theta = -21/29$.**

Example 12: Identity 4

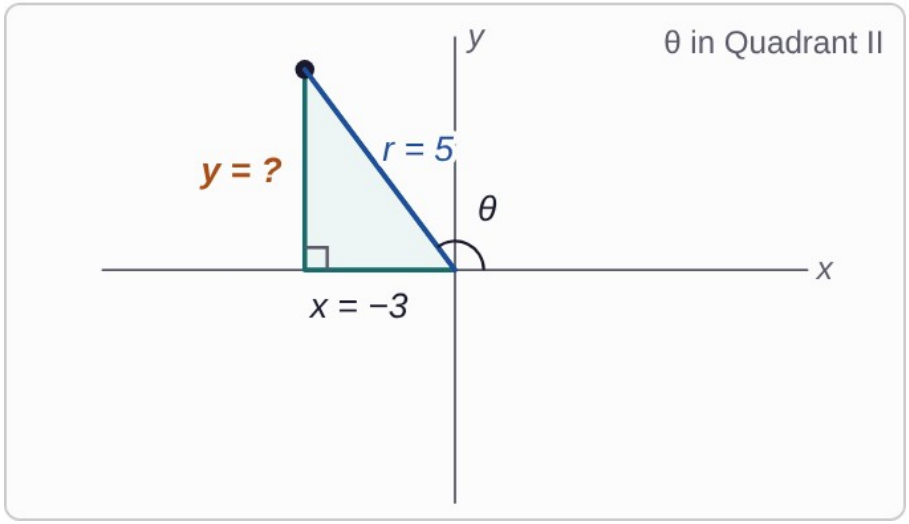


Figure: Coordinate axes with angle theta in Quadrant II. The terminal side goes from the origin up and to the left, with length r = 5. The horizontal leg along the x-axis is labeled x = negative 3. The vertical leg, from the x-axis up to the end of the terminal side, is labeled y = question mark.

$$\cos\theta = -3/5$$

Question

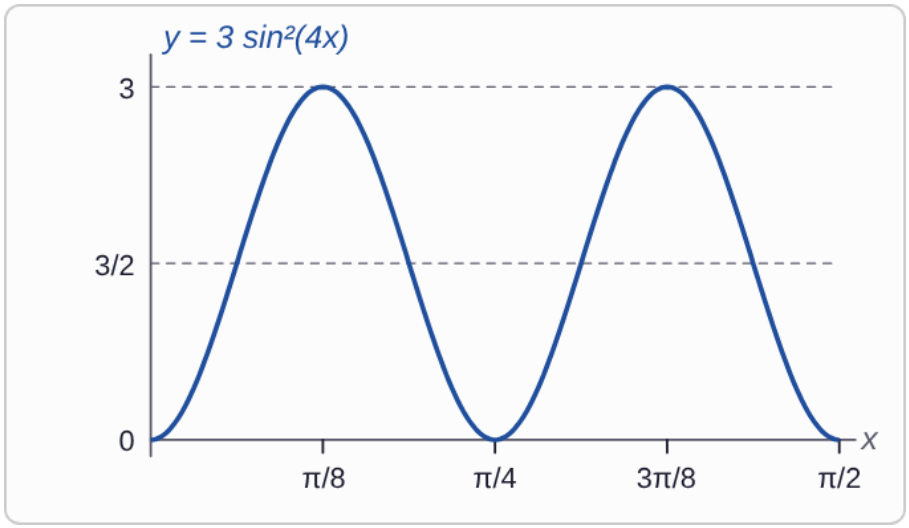
Angle θ is in Quadrant II. What is $\sin\theta$?

- **A.** $-4/5$
- **B.** $16/25$
- **C.** $2/5$
- **D.** $4/5$

Answer and reasoning

$\sin^2\theta = 1 - 9/25 = 16/25$. Stopping there gives $16/25$, but that is $\sin^2\theta$: take the square root, $\pm 4/5$. In Quadrant II, y is above zero, so the sign is not the same as the given cosine's. The answer $2/5$ subtracts $5 - 3$. **$\sin\theta = 4/5$.**

Example 13: Reduce 1



Plotted data: The graph of y = 3 sin squared of 4x for x from 0 to pi/2. The graph starts at 0 when x = 0, rises to 3 at x = pi/8, returns to 0 at x = pi/4, rises to 3 at x = 3pi/8, and returns to 0 at x = pi/2. Dashed lines mark y = 3/2 and y = 3. The graph is a wave that never goes below 0.

$$3\sin^2(4x)$$

Question

Which expression is equal to this one? The graph is a way to check.

- **A.** $3/2-3/2\cos(8x)$
- **B.** $3/2+3/2\cos(8x)$
- **C.** $3/2-3/2\cos(4x)$
- **D.** $1/2-3/2\cos(8x)$

Answer and reasoning

$\sin^2 u=(1-\cos(2u))/2$ with $u=4x$: $3\cdot(1-\cos(8x))/2=3/2-3/2\cos(8x)$. Check with the graph: at $x=0$ this gives $3/2-3/2=0$, and the midline is $y=3/2$. The plus version is the cosine formula and would start at 3. The $\cos(4x)$ version forgets to double the angle, and $1/2-\ldots$ multiplies only one term by 3. ***3/2−3/2 cos(8x).***

Example 14: Reduce 2

$$5\cos^2(2x)$$

Question

Which expression is equal to this one?

- **A.** $5/2-5/2\cos(4x)$
- **B.** $5/2+5/2\cos(2x)$
- **C.** $5/2+5/2\cos(4x)$
- **D.** $5+5\cos(4x)$

Answer and reasoning

$\cos^2 u=(1+\cos(2u))/2$ with $u=2x$: $5\cdot(1+\cos(4x))/2=5/2+5/2\cos(4x)$. Check at $x=0$: $5\cos^2 0=5$, and $5/2+5/2=5$. The minus version is the sine formula, the $\cos(2x)$ version does not double the angle, and $5+5\cos(4x)$ drops the division by 2. ***5/2+5/2* cos*(4x)*.**

Example 15: Reduce 3

$$2\sin^2(7x)$$

Question

Which expression is equal to this one?

- **A.** $1+\cos(14x)$
- **B.** $1-\cos(14x)$
- **C.** $1-\cos(7x)$
- **D.** $2-2\cos(14x)$

Answer and reasoning

$2\cdot(1-\cos(14x))/2=1-\cos(14x)$: the 2 cancels the 2 in the denominator. The plus version is the cosine formula, $1-\cos(7x)$ does not double the angle, and $2-2\cos(14x)$ forgets to divide by 2. ***1−cos(14x).***

Example 16: Reduce 4

$$4\cos^2(3x)$$

Question

Which expression is equal to this one?

- **A.** $2-2\cos(6x)$
- **B.** $2+2\cos(3x)$
- **C.** $2+2\cos(9x)$
- **D.** $2+2\cos(6x)$

Answer and reasoning

$4\cdot(1+\cos(6x))/2=2+2\cos(6x)$. The angle doubles, $2\cdot 3x=6x$; the 3 is not squared, so $\cos(9x)$ is out. The minus version is the sine formula, and $\cos(3x)$ does not double. **$2+2\cos(6x)$.**

Example 17: Slip 1

$$\sin(5\pi/12)=\sin(\pi/4)+\sin(\pi/6)$$
$$=(\sqrt{2}+1)/2$$

Question

A student found the exact value of $\sin(5\pi/12)$ with this work. What is the mistake, if any?

A. There is no mistake.
B. The angles should be π/2 − π/12 instead of π/4 + π/6.
C. Sine does not split over a sum like that.
D. It should be sin(π/4) − sin(π/6) instead.

Answer and reasoning

$\sin(A+B)=\sin A\cos B+\cos A\sin B$, not $\sin A+\sin B$. The formula gives $(\sqrt{2})/2\cdot(\sqrt{3})/2+(\sqrt{2})/2\cdot 1/2=(\sqrt{6}+\sqrt{2})/4\approx 0.966$. The student's answer is about 1.207, more than 1, which no sine can be. The angles $\pi/4+\pi/6$ were fine. **The correct value is $(\sqrt{6}+\sqrt{2})/4$.**

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Example 18: Slip 2

$$\sin^2\theta = 1-(20/29)^2 = 441/841$$
$$\sin\theta = -21/29$$

Question

Given $\cos\theta = 20/29$ with θ in Quadrant IV, a student found $\sin\theta$ with this work. What is the mistake, if any?

A. There is no mistake.
B. The sine should be 21/29, without the minus sign.
C. The answer should stay as 441/841.
D. It should be 1 − 20/29, so the sine is 9/29.

Answer and reasoning

$1-400/841 = 441/841$, and its square root is $21/29$. In Quadrant IV, y is below zero, so the minus sign is right. **There is no mistake.**

Example 19: Slip 3

$$\cos^2\theta = 1 - 49/625 = 576/625$$
$$\cos\theta = 24/25$$

Question

Given $\sin\theta = 7/25$ with θ in Quadrant II, a student found $\cos\theta$ with this work. What is the mistake, if any?

A. The cosine should be 18/25, from 25 − 7.
B. In Quadrant II the cosine is −24/25.
C. There is no mistake.
D. It should be 1 + 49/625, which makes the cosine more than 1.

Answer and reasoning

The square root gives $\pm 24/25$, and the student kept the wrong one. In Quadrant II, x is to the left of the origin, so the cosine is below zero. **$\cos\theta = -24/25$.**

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Example 20: Slip 4

$$\cos(3\pi/8) = 1/2 \cdot \cos(3\pi/4) = -(\sqrt{2})/4$$

Question

A student found the exact value of $\cos(3\pi/8)$ with this work. What is the mistake, if any?

A. There is no mistake.
B. It should be $2\cos(3\pi/4)$, which is $-\sqrt{2}$.
C. The answer should be above zero, so it is $\sqrt{2}/4$.
D. Halving the angle does not halve the cosine.

Answer and reasoning

The half-angle formula is $\cos(\theta/2) = \pm\sqrt{((1+\cos\theta)/2)}$. With $\theta = 3\pi/4$, $\cos\theta = -(\sqrt{2})/2$, and $3\pi/8$ is in Quadrant I, so take the positive root: $\sqrt{((1-\sqrt{2}/2)/2)} = (\sqrt{(2-\sqrt{2})})/2 \approx 0.383$. A quick check: an angle in Quadrant I cannot have a cosine below zero. Fixing only the sign, to $(\sqrt{2})/4$, still halves the cosine. **The correct value is $(\sqrt{(2-\sqrt{2})})/2$.**

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Example 21: Slip 5

$$4\sin^2(3x)=2-2\cos(3x)$$

Question

A student rewrote this expression with only first powers. What is the mistake, if any?

- **A.** The minus should be a plus: 2 + 2cos(3x).
- **B.** The angle should double, so it is 2 − 2cos(6x).
- **C.** There is no mistake.
- **D.** It should be 4 − 4cos(3x), keeping the 4 on both terms.

Answer and reasoning

$\sin^2 u=(1-\cos(2u))/2$: the angle inside the cosine is $2u$. With $u=3x$, that is $6x$, so $4\sin^2(3x)=2-2\cos(6x)$. Check at $x=\pi/6$: $4\sin^2\pi/2=4$, and $2-2\cos\pi=4$, but the student's version gives 2. ***2−2 cos(6x).***